

Experiment

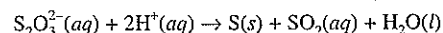
33

Effect of reactant concentration on reaction rate

Background

Different chemical reactions occur at different rates. Some reactions, such as many neutralisation and precipitation reactions, occur very rapidly. Other reactions, such as the air oxidation of vitamin C and the rusting of iron, are relatively slow. In this experiment you will investigate how one factor can affect the rate of reaction.

The reaction between sodium thiosulfate and hydrochloric acid can be represented by the equation



The time taken for the precipitate of sulfur to obscure a cross marked on a piece of paper placed under the reaction flask is used as a measure of the reaction rate. The less time taken for the cross to 'disappear', the faster is the rate of the reaction. In this activity you will investigate the rate of this reaction as a function of the sodium thiosulfate concentration.

Equipment required

Sodium thiosulfate solution $[\text{Na}_2\text{S}_2\text{O}_3]$ 0.25 mol L⁻¹ (150 mL)

Hydrochloric acid $[\text{HCl}]$ 2 mol L⁻¹ (30 mL)

Distilled water

Graduated cylinders (10 mL, 25 mL and 100 mL)

Conical flasks (two 100 mL)

Stop clock

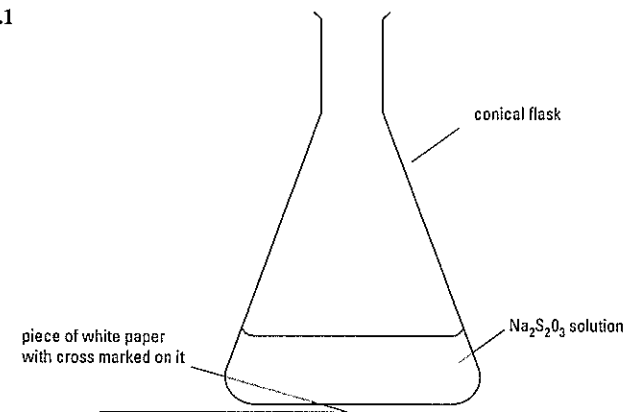
Procedure

#1 Copy the table below into your laboratory notebook and record your results directly into it.

Vol. of 0.25 mol L ⁻¹ Na ₂ S ₂ O ₃ (mL)	Vol. of H ₂ O added (mL)	Total volume after mixing (mL)	Concentration of Na ₂ S ₂ O ₃ on mixing (mol L ⁻¹)	Time for cross to disappear (seconds)	$\frac{1}{\text{time}}$ (s ⁻¹)
45	0	50	0.225	25	
35	10	50			
25	20	50			
15	30	50			
5	40	50			

- #2 Place 45 mL of 0.25 mol L⁻¹ Na₂S₂O₃ in a 100 mL conical flask. Put the flask over a cross marked on a piece of white paper as shown in Figure 33.1.
- #3 Add 5 mL of 2 mol L⁻¹ HCl and briefly agitate to ensure mixing of the reactants. Start a stop clock at the moment of addition.
- #4 Note and record the time taken for the cross to 'disappear' when it is viewed through the solution from directly overhead. The formation of solid sulfur causes the cross to be obscured.
- #5 Repeat the experiment using various sodium thiosulfate concentrations, made up as indicated in the table.

Fig. 33.1



Processing of results, and questions

- 1 Calculate 1/time for each experiment and enter the results into the table.
- 2 Plot a graph of 1/time (a measure of the reaction rate) against sodium thiosulfate concentration.
- 3 What effect does the concentration of sodium thiosulfate have on the reaction rate?
- 4 If the concentration of sodium thiosulfate is doubled, what happens to the rate of the reaction?

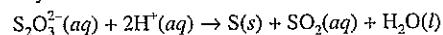
Experiment

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Effect of temperature
on reaction rate

Background

The purpose of this experiment is to investigate the effect of temperature on reaction rate. The reaction observed is the same as in the previous experiment, between sodium thiosulfate and hydrochloric acid.



The time taken for the precipitate of sulfur to obscure a cross marked on a piece of paper placed under the reaction flask is used as a measure of the reaction rate. The less time taken for the cross to 'disappear', the faster is the rate of reaction.

Equipment required

Sodium thiosulfate solution $[\text{Na}_2\text{S}_2\text{O}_3]$ 0.25 mol L⁻¹ (80 mL)

Hydrochloric acid $[\text{HCl}]$ 2 mol L⁻¹ (30 mL)

Distilled water

Conical flask (100 mL)

Test tube

Thermometer (-10–110 °C)

Stop clock

Graduated cylinders (10 mL, 25 mL and 100 mL)

Beaker (500 mL) to be used as a water bath

Bunsen, tripod and gauze mat

Procedure

- #1 Place 15 mL of 0.25 mol L⁻¹ Na₂S₂O₃ and 30 mL of water into a 100 mL conical flask.
- #2 Measure 5 mL of 2 mol L⁻¹ HCl into a test tube.
- #3 Heat the solutions separately in a water bath until they are slightly above 20 °C.
- #4 Put the flask over a cross marked on a piece of white paper and add the acid to the thiosulfate solution. Start a stop clock at the moment of addition.
- #5 Note the time taken for the cross to 'disappear' and record the average temperature during the reaction.
- #6 Repeat the procedure for temperatures of about 30, 40, 50 and 60 °C. In each case you will need to measure the average temperature for the reaction.
- #7 Construct a table showing the temperature, times taken for the cross to disappear, and 1/time.

Processing of results, and questions

- 1 Plot a graph of 1/time against temperature.
- 2 What effect does increasing the temperature have on the reaction rate?
- 3 Suggest two factors that contribute to the change in reaction rate with temperature.

Experiment

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Effect of catalysts on reaction rate

Background

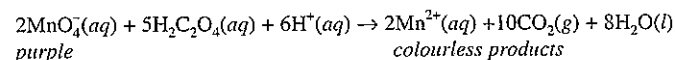
Catalysts are substances that increase the rate of chemical reactions but are not consumed in these reactions. Catalysts therefore take part in chemical reactions but are regenerated at the end of the reactions.

Catalysts are important in many industrial processes such as the manufacture of ammonia and sulfuric acid and in biochemical processes which are catalysed by enzymes.

In this experiment the effect of catalysts on the rate of three different reactions will be observed. The three reactions are:

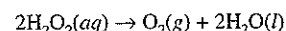
- 1 The reaction of permanganate ion with oxalic acid

In this reaction the purple permanganate ion reacts with oxalic acid to form carbon dioxide and almost colourless manganese(II) ion. The equation is



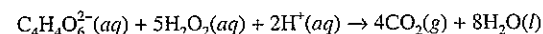
- 2 The decomposition of hydrogen peroxide

Hydrogen peroxide is unstable and slowly decomposes over time according to the equation



- 3 The reaction of tartrate ion with hydrogen peroxide

In this reaction hydrogen peroxide oxidises tartrate ion to carbon dioxide according to the equation



Equipment required

Sulfuric acid [H_2SO_4] 2 mol L^{-1} (60 mL)
 Potassium permanganate solution [KMnO_4] 0.02 mol L^{-1} (10 mL)
 Saturated solution of oxalic acid [$\text{H}_2\text{C}_2\text{O}_4$] (10 mL)
 Saturated solution of manganese(II) sulfate [MnSO_4] (1 mL)
 Fresh hydrogen peroxide solution [H_2O_2] 6% (50 mL)
 Potassium sodium tartrate [$\text{KNaC}_4\text{H}_4\text{O}_6$] (about 1 g)
 Manganese(IV) oxide [MnO_2] (a few grains)
 Cobalt(II) chloride-6-water solution [$\text{CoCl}_2 \cdot 6\text{H}_2\text{O}$] 10% solution (1 mL)
 Distilled water
 Beakers (two 250 mL and two 100 mL)
 Thermometer (-10–110 °C)
 Plastic teaspoon

Procedure

A Reaction of Permanganate Ion with Oxalic Acid

Safety Note

- Oxalic acid is poisonous and should not be allowed to come in contact with your skin.
- Handle the $\text{H}_2\text{C}_2\text{O}_4$ solution carefully.

#1 In each of two 250 mL beakers place 25 mL of 2 mol L^{-1} H_2SO_4 , 125 mL of distilled water, 3 mL of 0.02 mol L^{-1} KMnO_4 and 3 mL of saturated oxalic acid solution. Is there any evidence of reaction?

#2 To one of the beakers add about 1 mL of saturated manganese(II) sulfate solution and record your observations.

B Decomposition of Hydrogen Peroxide

Safety Note

- Hydrogen peroxide can cause burns if it comes in contact with your skin.
- Handle the H_2O_2 solution carefully.
- Add only a few grains of manganese(IV) oxide to the H_2O_2 otherwise the reaction will be too vigorous.

#1 Place about 20 mL of approximately 6% hydrogen peroxide solution into each of two 100 mL beakers. Is there any evidence of reaction?

#2 Into one of the beakers add a few grains of manganese(IV) oxide and record your observations.

C Reaction of Tartrate Ion with Hydrogen Peroxide

#1 Dissolve 1 g of potassium sodium tartrate in 10 mL of water and add 10 mL of 6% hydrogen peroxide.

#2 Heat the solution to 65 °C and remove the bunsen. Is there any evidence of reaction?

#3 Add about 1 mL of 10% cobalt(II) chloride solution and carefully observe any changes.

Processing of results, and questions

- 1 What species was acting as the catalyst in each reaction?
- 2 In reaction C, what evidence is there that the catalyst was not consumed in the reaction? Is there any evidence that the catalyst took an active part in the reaction?